

UptimeLog Project (Simple Idea)

Goal

Create a small Linux CLI tool written in C++ that automatically tracks how long a Linux system stays running (uptime) and keeps a history of it.

How It Works

1. A background service starts automatically when Linux boots.
 2. The service reads system uptime from `/proc/uptime`.
 3. It calculates when the system booted.
 4. It saves the session information into a history file.
 5. The CLI tool lets the user view and analyze the history.
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Two Parts of the Tool

1. uptimeLogd (Daemon)

- Runs in the background
- Starts automatically when the system boots
- Reads system uptime
- Records uptime sessions

2. uptimeLog (CLI Tool)

- Shows current uptime
 - Shows session history
 - Shows statistics
 - Allows adding notes
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Where Data Is Stored

`~/.local/share/uptimeLog/history.json`

Example stored session:

```
{
  "id": 1,
  "boot_time": "2026-03-11T10:00:00",
```

```
"shutdown_time": "2026-03-11T18:00:00",  
"uptime_seconds": 28800,  
"note": ""  
}
```

Example CLI Commands

```
uptimelog status  
uptimelog history  
uptimelog stats  
uptimelog note 3 "Power outage"
```

Development Roadmap

```
Phase 1 → read /proc/uptime  
Phase 2 → build simple CLI tool  
Phase 3 → store uptime history  
Phase 4 → create background daemon  
Phase 5 → add systemd service  
Phase 6 → add stats & notes
```

What You Learn From This Project

- Linux system files (/proc)
- C++ CLI programming
- Background services (daemons)
- Structured data storage (JSON)
- Building real Linux utilities

Final Result

A lightweight Linux utility that automatically records system uptime sessions and lets users analyze the reliability and usage of their machine through a simple CLI interface.